

Supplementary Material

Deep multi-agent reasoning in hyperbolic discs

S1. Learned parameters for the single-agent reasoning task

The learned coupling matrix is:

$$K = \begin{bmatrix} 0 & 208.76 & 0 \\ 208.76 & 0 & 134.91 \\ 0 & 134.91 & 0 \end{bmatrix}$$

The learned frequency is:

$$\omega = 5.26$$

The initial states are:

$$z(0) = \begin{bmatrix} 0.000315 - 0.007645j \\ -0.000084 - 0.000113j \\ -0.001395 + 0.000665j \end{bmatrix}$$

The final states are:

$$z(T) = \begin{bmatrix} -0.898472 - 0.438978j \\ -0.898119 - 0.439954j \\ -0.898403 - 0.439160j \end{bmatrix}$$

S2. Structure of the generated dataset

This section provides an example of the JSON representation used to define the hierarchical decision structure and conditional probabilities.

```
{
  "scenario": "Multi-Agent Layered Decision Tree via Maximum Entropy",
  "depth": 4,
  "branching_factor": 3,
  "conditional_probabilities": [
    {
      "condition": { "father": "F1_pub", "mother": "M1_entertainment" },
      "effects": {
        "son.S1_home": 0.05,
        "son.S1_outdoors": 0.90,
        "son.S1_travel": 0.05
      }
    },
    {
      "condition": { "father": "F1_pub", "mother": "M1_home" },
      "effects": {
        "son.S1_home": 0.3333333333333333,
        "son.S1_outdoors": 0.3333333333333333,
        "son.S1_travel": 0.3333333333333333
      }
    }
  ]
}
```

```

    }
  }
]
}

```

S3. Hierarchical state architecture

Table S1 presents the complete hierarchical state structure for the three agents.

Table S1: Hierarchical state structure of the three agents (Father, Mother, Son)

Agent	Level 1 (L_1)	Level 2 (L_2)	Level 3 / Terminal Choices (L_3)
(A_1)	F1_home	F2_movie	F3_movie_lore, F3_popcorn_guy, F3_critic
		F2_book	F3_read_history, F3_read_sci_book, F3_read_novel
		F2_rest	F3_couch_nap, F3_garden_work, F3_meditate
	F1_pub	F2_beer_tasting	F3_order_ipa, F3_order_lager, F3_order_craft
		F2_pub_games	F3_play_billiards, F3_play_cards, F3_deep_talk
		F2_sports_screening	F3_watch_match, F3_check_stats, F3_cheer_team
	F1_travel	F2_seaside	F3_sailing, F3_deep_diving, F3_swim_laps
		F2_mountains	F3_heavy_hiking, F3_rock_climbing, F3_chop_wood
		F2_city	F3_history_museum, F3_architecture, F3_rock_concert
(A_2)	M1_home	M2_movie	M3_movie_plot, M3_emotional_scenes, M3_enjoy_snacks
		M2_book	M3_read_poetry, M3_read_psychology, M3_read_biography
		M2_rest	M3_do_yoga, M3_herbal_tea, M3_bake_cookies
	M1_entertainment	M2_shopping	M3_art_print_hunting, M3_indie_perfume, M3_vintage_finds
		M2_workshop	M3_pottery_wheel, M3_floral_design, M3_pastry_baking
		M2_cafe_relax	M3_botanical_patio, M3_read_magazine, M3_patio_espresso
	M1_travel	M2_seaside	M3_sunbathing, M3_seafood_dinner, M3_beach_spa
		M2_mountains	M3_nature_photography, M3_alpine_resort, M3_scenic_views
		M2_city	M3_art_gallery, M3_fashion_boutique, M3_opera_theatre
(A_3)	S1_home	S2_movie	S3_movie_effects, S3_watch_cartoons, S3_fall_asleep
		S2_book	S3_read_comics, S3_read_fantasy, S3_read_manga
		S2_rest	S3_play_video_games, S3_build_lego, S3_sleep_late
	S1_outdoors	S2_arcade	S3_play_games, S3_win_tickets, S3_eat_pizza
		S2_park	S3_ride_bmx, S3_see_graffiti, S3_skate_halfpipe
		S2_walk	S3_ride_scooter, S3_ride_bicycle, S3_walk_the_dog
	S1_travel	S2_seaside	S3_swimming, S3_water_slides, S3_sandcastles
		S2_mountains	S3_sledding, S3_camping, S3_hiking
		S2_city	S3_visit_toy_shop, S3_visit_zoo, S3_eat_fast_food

S4. Example story-generation scenarios

This section demonstrates a coherent and divergent narratives for the multi-agent reasoning task.

S4.1. Coherent narrative

The agents make compatible decisions across the three layers. The resulting path is:

1. **Layer 1:** Father selects F1_travel, followed by Mother selecting M1_travel and Son selecting S1_travel.
2. **Layer 2:** The agents remain within the same context by selecting F2_mountains, M2_mountains, and S2_mountains.
3. **Layer 3:** The agents select compatible terminal actions, including F3_chop_wood, M3_scenic_views, and S3_camping.

Resulting narrative: The shared decision was to pack the bags and go on a travel getaway. They traveled up into the fresh, crisp air of the mountains. Father gathered fallen logs around the site and began to chop wood for later. Mother found an elevated lookout point to quietly admire the endless scenic views. The Son loved the outdoors, happily setting up his sleeping bag for true tent camping.

S4.2. Divergent narrative

The agents select incompatible contexts. The resulting path is:

1. **Layer 1:** Father selects F1_pub, while Mother selects M1_home. The Son therefore selects S1_travel with uniform probability $P = 1/3$.
2. **Layer 2:** Father selects F2_pub_games, while Mother selects M2_book. The Son selects S2_mountains from a uniform distribution over the available states.
3. **Layer 3:** Father selects F3_play_cards and Mother selects M3_read_poetry, while the Son randomly selects S3_sledding.

Resulting narrative: The father is playing cards in a pub, the mother is reading poetry at home, and the son is sledding in the mountains. The resulting path lacks a common context.

S5. Learned parameters for the multi-agent reasoning task

The learned coupling matrix is:

$$K = \begin{bmatrix} 0 & 16.96 & 0 & -9.69 & 0 & 0 & -54.04 & 0 & 0 \\ 16.957 & 0 & 124.82 & 0 & 59.36 & 0 & 0 & -64.716 & 0 \\ 0 & 124.82 & 0 & 0 & 0 & -82.45 & 0 & 0 & 8.09 \\ -9.69 & 0 & 0 & 0 & 86.65 & 0 & -93.68 & 0 & 0 \\ 0 & 59.36 & 0 & 86.65 & 0 & 55.83 & 0 & 61.83 & 0 \\ 0 & 0 & -82.45 & 0 & 55.83 & 0 & 0 & 0 & 59.34 \\ -54.04 & 0 & 0 & -93.68 & 0 & 0 & 0 & 64.11 & 0 \\ 0 & -64.71 & 0 & 0 & 61.83 & 0 & 64.11 & 0 & 134.49 \\ 0 & 0 & 8.09 & 0 & 0 & 59.34 & 0 & 134.49 & 0 \end{bmatrix}$$

The learned frequency is:

$$\omega = 92.71$$

The initial states are:

$$z(0) = \begin{bmatrix} -0.002471 + 0.007392j \\ 0.018609 - 0.053924j \\ 0.000664 + 0.001299j \\ 0.002359 + 0.000227j \\ -0.014091 - 0.009191j \\ 0.000849 + 0.000084j \\ -0.000479 + 0.000881j \\ -0.004972 - 0.001342j \\ -0.007021 - 0.003103j \end{bmatrix}$$

The final states are:

$$z(T) = \begin{bmatrix} -0.451681 + 0.891652j \\ -0.567424 + 0.823437j \\ -0.767154 + 0.641436j \\ 0.287887 + 0.957584j \\ 0.750346 + 0.660874j \\ 0.968774 - 0.247753j \\ 0.297651 - 0.954658j \\ 0.866143 - 0.499891j \\ 0.909331 - 0.416025j \end{bmatrix}$$